

CARBON MARKETS CCUS AND DIRECT AIR CAPTURE

CARBON-UNIT IDENTITY FORK → MARKET ARCHITECTURE MATRIX → CLAIM-TYPE LADDER → INTEGRITY GATE → CREDIT LIFECYCLE RAIL → ACCOUNTING-RISK FIREWALL

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g6 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 CARBON-UNIT IDENTITY FORK

CARBON-UNIT IDENTITY FORK PERMISSION UNIT INSIDE A COMPLIANCE CONSTRAINT VERIFIED OUTCOME AGAINST AN APPROVED BASELINE ALLOCATION OR AUCTION VERSUS METHODOLOGY AND ISSUANCE SURRENDER OR RETIREMENT UNDER THE APPLICABLE RULE

TRADABILITY DOES NOT ERASE DIFFERENT UNIT ORIGINS MUST REMEMBER CARBON PRICING AND MARKETS ASSIGN TRADABLE OR FISCAL INCENTIVES

ALLOWANCE → permission unit inside a compliance constraint

CREDIT → verified outcome against an approved baseline

ORIGIN → allocation or auction versus methodology and issuance

USE → surrender or retirement under the applicable rule

CORE CLAIM

- ALLOWANCE → permission unit inside a compliance constraint
- CREDIT → verified outcome against an approved baseline

MECHANISM

- ORIGIN → allocation or auction versus methodology and issuance
- USE → surrender or retirement under the applicable rule

CONSEQUENCE / LIMIT

- TRAP → tradability does not erase different unit origins
- MUST REMEMBER: Carbon pricing and markets assign tradable or fiscal incentives to...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → tradability does not erase different unit origins.

01

STAGE 01 MARKET ARCHITECTURE MATRIX

MARKET ARCHITECTURE MATRIX CAP AND TRADE ABSOLUTE COVERED-EMISSIONS CEILING BASELINE AND CREDIT PERFORMANCE AGAINST TARGET OR BASELINE LEGAL OBLIGATION AND SURRENDER REQUIREMENT NON-MANDATED PURCHASE OR CLAIM

WHAT CONSTRAINS EMISSIONS AND WHO MUST ACT

CORE CLAIM

CAP AND TRADE → absolute covered-emissions ceiling
BASELINE AND CREDIT → performance against target or baseline

MECHANISM

COMPLIANCE → legal obligation and surrender requirement
VOLUNTARY → non-mandated purchase or claim

CONSEQUENCE / LIMIT

QUESTION → what constrains emissions and who must act
CAP AND TRADE → absolute covered-emissions ceiling

CORE CLAIM

- CAP AND TRADE → absolute covered-emissions ceiling
- BASELINE AND CREDIT → performance against target or baseline

MECHANISM

- COMPLIANCE → legal obligation and surrender requirement
- VOLUNTARY → non-mandated purchase or claim

CONSEQUENCE / LIMIT

- QUESTION → what constrains emissions and who must act
- CAP AND TRADE → absolute covered-emissions ceiling

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: QUESTION → what constrains emissions and who must act.

02

STAGE 02 CLAIM-TYPE LADDER

CLAIM-TYPE LADDER COUNTERFACTUAL SOURCE DOES NOT ARISE EXISTING SOURCE EMITS LESS CARBON LEAVES AMBIENT ATMOSPHERE DURATION AND REVERSAL RISK DETERMINE DURABILITY NO MERGER EACH SUPPORTS A DIFFERENT CLIMATE CLAIM

AVOIDANCE → counterfactual source does not arise

REDUCTION → existing source emits less

REMOVAL → carbon leaves ambient atmosphere

STORAGE → duration and reversal risk determine durability

CORE CLAIM

- AVOIDANCE → counterfactual source does not arise
- REDUCTION → existing source emits less

MECHANISM

- REMOVAL → carbon leaves ambient atmosphere
- STORAGE → duration and reversal risk determine durability

CONSEQUENCE / LIMIT

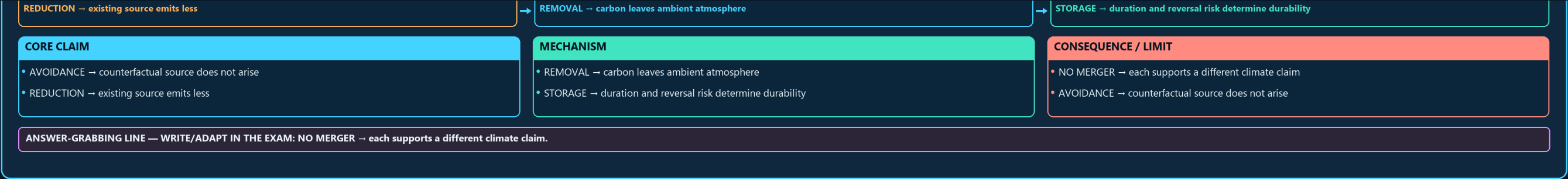
- NO MERGER → each supports a different climate claim
- AVOIDANCE → counterfactual source does not arise

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NO MERGER → each supports a different climate claim.

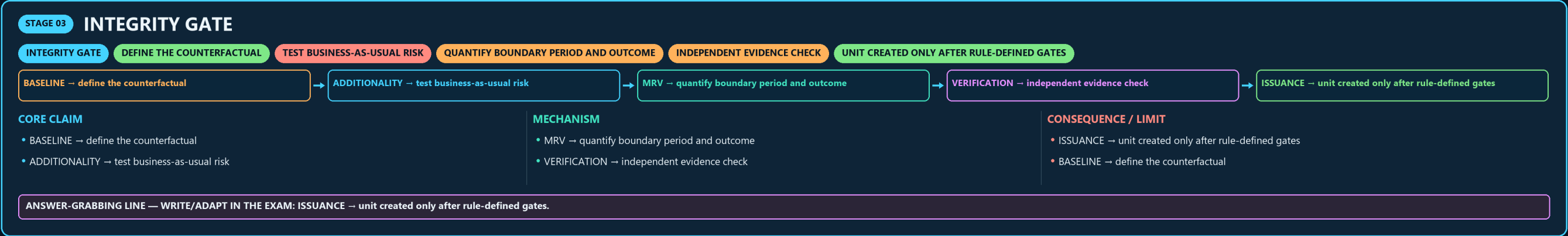
03

STAGE 03 INTEGRITY GATE

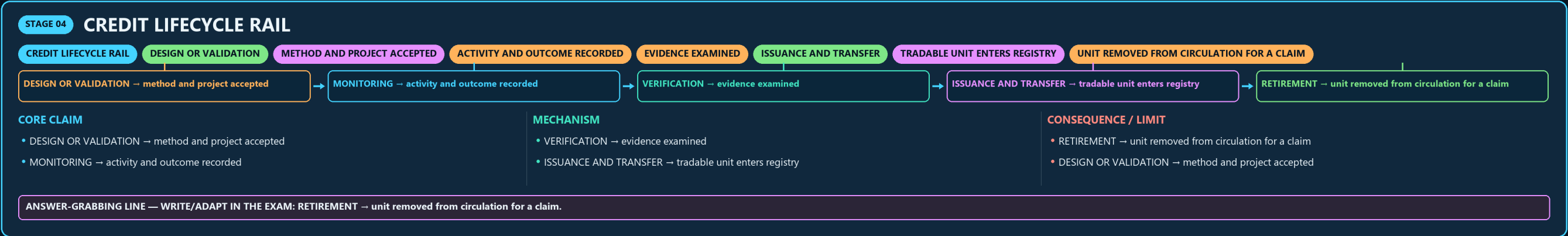
INTEGRITY GATE DEFINE THE COUNTERFACTUAL TEST BUSINESS-AS-USUAL RISK QUANTIFY BOUNDARY PERIOD AND OUTCOME INDEPENDENT EVIDENCE CHECK UNIT CREATED ONLY AFTER RULE-DEFINED GATES



03



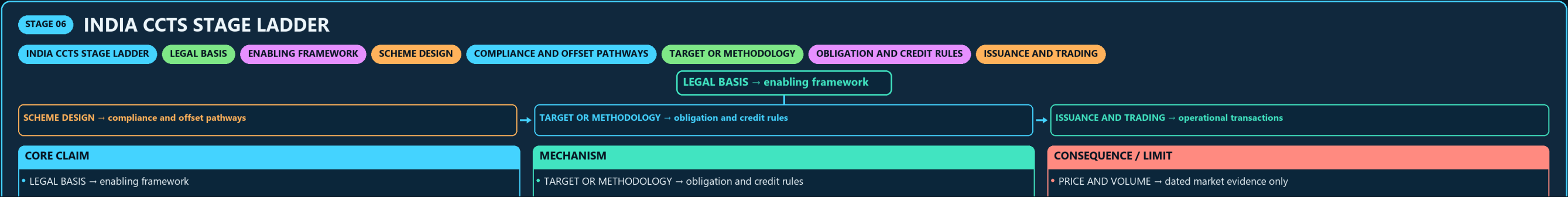
04



05



06



ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → adjustment is not issuance transfer or retirement.

06

STAGE 06 INDIA CCTS STAGE LADDER

INDIA CCTS STAGE LADDER

LEGAL BASIS

ENABLING FRAMEWORK

SCHEME DESIGN

COMPLIANCE AND OFFSET PATHWAYS

TARGET OR METHODOLOGY

OBLIGATION AND CREDIT RULES

ISSUANCE AND TRADING

LEGAL BASIS → enabling framework

SCHEME DESIGN → compliance and offset pathways

TARGET OR METHODOLOGY → obligation and credit rules

ISSUANCE AND TRADING → operational transactions

CORE CLAIM

- LEGAL BASIS → enabling framework
- SCHEME DESIGN → compliance and offset pathways

MECHANISM

- TARGET OR METHODOLOGY → obligation and credit rules
- ISSUANCE AND TRADING → operational transactions

CONSEQUENCE / LIMIT

- PRICE AND VOLUME → dated market evidence only
- LEGAL BASIS → enabling framework

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PRICE AND VOLUME → dated market evidence only.

07

STAGE 07 CCUS CHAIN

CCUS CHAIN

CONCENTRATED CARBON DIOXIDE STREAM

CAPTURE AND CONDITIONING

SEPARATION AND PREPARATION

PIPELINE SHIP ROAD OR OTHER ROUTE

UTILISATION OR STORAGE

DESTINATION DETERMINES DURABILITY

MONITORING AND CLOSURE

SOURCE → concentrated carbon dioxide stream

CAPTURE AND CONDITIONING → separation and preparation

TRANSPORT → pipeline ship road or other route

UTILISATION OR STORAGE → destination determines durability

MONITORING AND CLOSURE → permanence and responsibility

CORE CLAIM

- SOURCE → concentrated carbon dioxide stream
- CAPTURE AND CONDITIONING → separation and preparation

MECHANISM

- TRANSPORT → pipeline ship road or other route
- UTILISATION OR STORAGE → destination determines durability

CONSEQUENCE / LIMIT

- MONITORING AND CLOSURE → permanence and responsibility
- SOURCE → concentrated carbon dioxide stream

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRANSPORT → pipeline ship road or other route.

08

STAGE 08 UTILISATION-STORAGE TEST

UTILISATION-STORAGE TEST

PRODUCT USE

HOW LONG IS CARBON RETAINED

PROCESS USE

DOES LATER RELEASE OCCUR

GEOLOGICAL STORAGE

CONTAINMENT AND MONITORING

REVERSAL RISK AND REMEDIATION

PRODUCT USE → how long is carbon retained

PROCESS USE → does later release occur

GEOLOGICAL STORAGE → containment and monitoring

LEAKAGE → reversal risk and remediation

CORE CLAIM

- PRODUCT USE → how long is carbon retained
- PROCESS USE → does later release occur

MECHANISM

- GEOLOGICAL STORAGE → containment and monitoring
- LEAKAGE → reversal risk and remediation

CONSEQUENCE / LIMIT

- NET BENEFIT → full lifecycle accounting required
- PRODUCT USE → how long is carbon retained

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PRODUCT USE → how long is carbon retained.

09

STAGE 09 CAPTURE TECHNOLOGY COMPARISON

CAPTURE TECHNOLOGY COMPARISON

POINT SOURCE CCUS

CARBON INTERCEPTED BEFORE RELEASE

CARBON SEPARATED FROM AMBIENT AIR

BIOGENIC CARBON PLUS CAPTURE AND STORAGE

REMOVAL TEST

ATMOSPHERIC ORIGIN PLUS DURABLE STORAGE

LIFECYCLE TEST

CORE CLAIM

POINT SOURCE CCUS → carbon intercepted before release

DAC → carbon separated from ambient air

MECHANISM

BECCS → biogenic carbon plus capture and storage

REMOVAL TEST → atmospheric origin plus durable storage

CONSEQUENCE / LIMIT

LIFECYCLE TEST → energy land transport and reversal

CLOSE DISTINCTION: Allowance is not offset, avoidance is not removal, capture is not...

CORE CLAIM

- POINT SOURCE CCUS → carbon intercepted before release
- DAC → carbon separated from ambient air

MECHANISM

- BECCS → biogenic carbon plus capture and storage
- REMOVAL TEST → atmospheric origin plus durable storage

CONSEQUENCE / LIMIT

- LIFECYCLE TEST → energy land transport and reversal
- CLOSE DISTINCTION: Allowance is not offset, avoidance is not removal, capture is not...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Allowance is not offset, avoidance is not removal, capture is not.

09

STAGE 09

CAPTURE TECHNOLOGY COMPARISON

CAPTURE TECHNOLOGY COMPARISON

POINT SOURCE CCUS

CARBON INTERCEPTED BEFORE RELEASE

CARBON SEPARATED FROM AMBIENT AIR

BIOGENIC CARBON PLUS CAPTURE AND STORAGE

REMOVAL TEST

ATMOSPHERIC ORIGIN PLUS DURABLE STORAGE

LIFECYCLE TEST

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
POINT SOURCE CCUS → carbon intercepted before release	BECCS → biogenic carbon plus capture and storage	LIFECYCLE TEST → energy land transport and reversal
DAC → carbon separated from ambient air	REMOVAL TEST → atmospheric origin plus durable storage	CLOSE DISTINCTION: Allowance is not offset, avoidance is not removal, capture is not...

CORE CLAIM

• POINT SOURCE CCUS → carbon intercepted before release

• DAC → carbon separated from ambient air

MECHANISM

• BECCS → biogenic carbon plus capture and storage

• REMOVAL TEST → atmospheric origin plus durable storage

CONSEQUENCE / LIMIT

• LIFECYCLE TEST → energy land transport and reversal

• CLOSE DISTINCTION: Allowance is not offset, avoidance is not removal, capture is not...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Allowance is not offset, avoidance is not removal, capture is not.

10

STAGE 10

NET-ZERO MITIGATION HIERARCHY

NET-ZERO MITIGATION HIERARCHY

PREVENT UNNECESSARY EMISSIONS

DECARBONISE EXISTING ACTIVITIES

ADDRESS HARD-TO-ABATE POINT SOURCES

COUNTERBALANCE GENUINELY RESIDUAL EMISSIONS

NET BALANCE NEEDS ONE BOUNDARY AND PERIOD

EVIDENCE LIMIT

FIX SYSTEM BOUNDARY, BASELINE, UNIT

REDUCE → decarbonise existing activities

CAPTURE → address hard-to-abate point sources

REMOVE → counterbalance genuinely residual emissions

AVOID → prevent unnecessary emissions

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
• AVOID → prevent unnecessary emissions • REDUCE → decarbonise existing activities	• CAPTURE → address hard-to-abate point sources • REMOVE → counterbalance genuinely residual emissions	• VERIFY → net balance needs one boundary and period • EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Fix system boundary, baseline, unit,...

MECHANISM / VERDICT — EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Fix system boundary, baseline, unit,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Fix system boundary, baseline, unit,.

11

STAGE 11

CARBON INSTRUMENTS ANSWER SPINE

CARBON INSTRUMENTS ANSWER SPINE

UNIT MARKET AND CLAIM TYPE

BASELINE MRV ISSUANCE TRANSFER RETIREMENT

CAPTURE TRANSPORT UTILISATION OR STORAGE

PERMANENCE LEAKAGE LIFECYCLE AND DOUBLE COUNTING

CURRENT RULE PRICE VOLUME COST AND CAPACITY SOURCE

CLASSIFY → unit market and claim type

TRACE → baseline MRV issuance transfer retirement

FOLLOW → capture transport utilisation or storage

TEST → permanence leakage lifecycle and double counting

AUDIT → current rule price volume cost and capacity source

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
• CLASSIFY → unit market and claim type • TRACE → baseline MRV issuance transfer retirement	• FOLLOW → capture transport utilisation or storage • TEST → permanence leakage lifecycle and double counting	• AUDIT → current rule price volume cost and capacity source • CLASSIFY → unit market and claim type

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: AUDIT → current rule price volume cost and capacity source.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT

DAC'S FUNDAMENTAL TECHNICAL CHALLENGE IS THERMODYNAMIC

ATMOSPHERIC CO₂ CONCENTRATION

POLICY IMPLICATION

BECAUSE DAC DELIVERS GENUINE ATMOSPHERIC REMOVAL (UNLIKE

GLOBAL CCUS AND DAC DEPLOYMENT CAPACITY REMAINS SMALL RELATIVE

THE INDIAN CARBON MARKET'S SPECIFIC SECTORAL COVERAGE, TARGET-SETTING METHODOLOGY

MRV-SYSTEM ROBUSTNESS FOR BOTH INDIA'S DOMESTIC CARBON MARKET AND

OPTIONAL SOURCE DEPTH	ADVANCED DEBATE	LEGACY / NUANCE
• CAUTION: DAC's fundamental technical challenge is thermodynamic: atmospheric CO₂ concentration • CAUTION: Policy implication: because DAC delivers genuine atmospheric removal (unlike • CAUTION: Global CCUS and DAC deployment capacity remains small relative to the scale IPCC • CAUTION: The Indian Carbon Market's specific sectoral coverage, target-setting methodology and	• CAUTION: MRV-system robustness for both India's domestic carbon market and international • India's carbon market has a two-track design: a compliance mechanism (building on PAT, • Standard fossil-source CCUS is an emissions-avoidance technology; genuine atmospheric • DAC's higher cost/energy-intensity relative to CCUS stems from the much lower CO₂	• Net-zero frameworks require genuine removal technologies (DAC/BECCS/nature-based sinks) • NOT: All CCUS applications, including standard fossil-power-plant CCUS, deliver negative • NOT: India's carbon market is a single-track, purely compliance-based system. - It has a • NOT: DAC is currently deployed at a scale comparable to global CCUS capacity. - DAC remains

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.